

Task 1 — Linear Regression: California House Price Predictor



House Price Prediction using Linear Regression



Introduction

Machine Learning is becoming an important part of modern technology, especially in areas where predictions and data analysis are required. In this project, I built a House Price Prediction system using the California Housing dataset and the Linear Regression algorithm.

The purpose of this project was not only to train a machine learning model, but also to understand the complete workflow involved in a real ML project. Throughout this project, I learned how data is collected, explored, analyzed, visualized, and finally used for prediction.

This project includes:

- Loading and understanding the dataset
- Performing Exploratory Data Analysis (EDA)
- Visualizing data using graphs
- Splitting data into training and testing sets
- Training a Linear Regression model
- Evaluating model performance
- Saving the trained model for future use

The project helped me understand how machine learning models can use historical data to predict future values.



Dataset Description

The California Housing dataset was used in this project. This dataset contains information collected from different districts of California and is commonly used for regression-based machine learning tasks.

The dataset includes multiple features related to houses and population statistics, such as:

- Median Income
- House Age
- Average Rooms

- Average Bedrooms
- Population
- Occupancy
- Latitude
- Longitude

The target variable in this dataset is the Median House Value, which represents the average price of houses in a particular district.

Each row in the dataset represents information from one district, while each column represents a feature that may affect house prices.

The dataset is suitable for machine learning because it contains numerical values and meaningful relationships between features and house prices.

Exploratory Data Analysis (EDA)

Before training the machine learning model, I performed Exploratory Data Analysis (EDA) to better understand the dataset and identify important patterns.

The first step was checking the shape of the dataset to understand the number of rows and columns. Then I used functions such as `info()` and `describe()` to study data types, statistical summaries, and feature distributions.

I also checked for missing values using `isnull().sum()`. Fortunately, the dataset did not contain missing values, which made preprocessing easier.

A correlation heatmap was created to understand relationships between features. The heatmap showed that Median Income has a strong positive correlation with house prices. This means that districts with higher median income generally have higher house values.

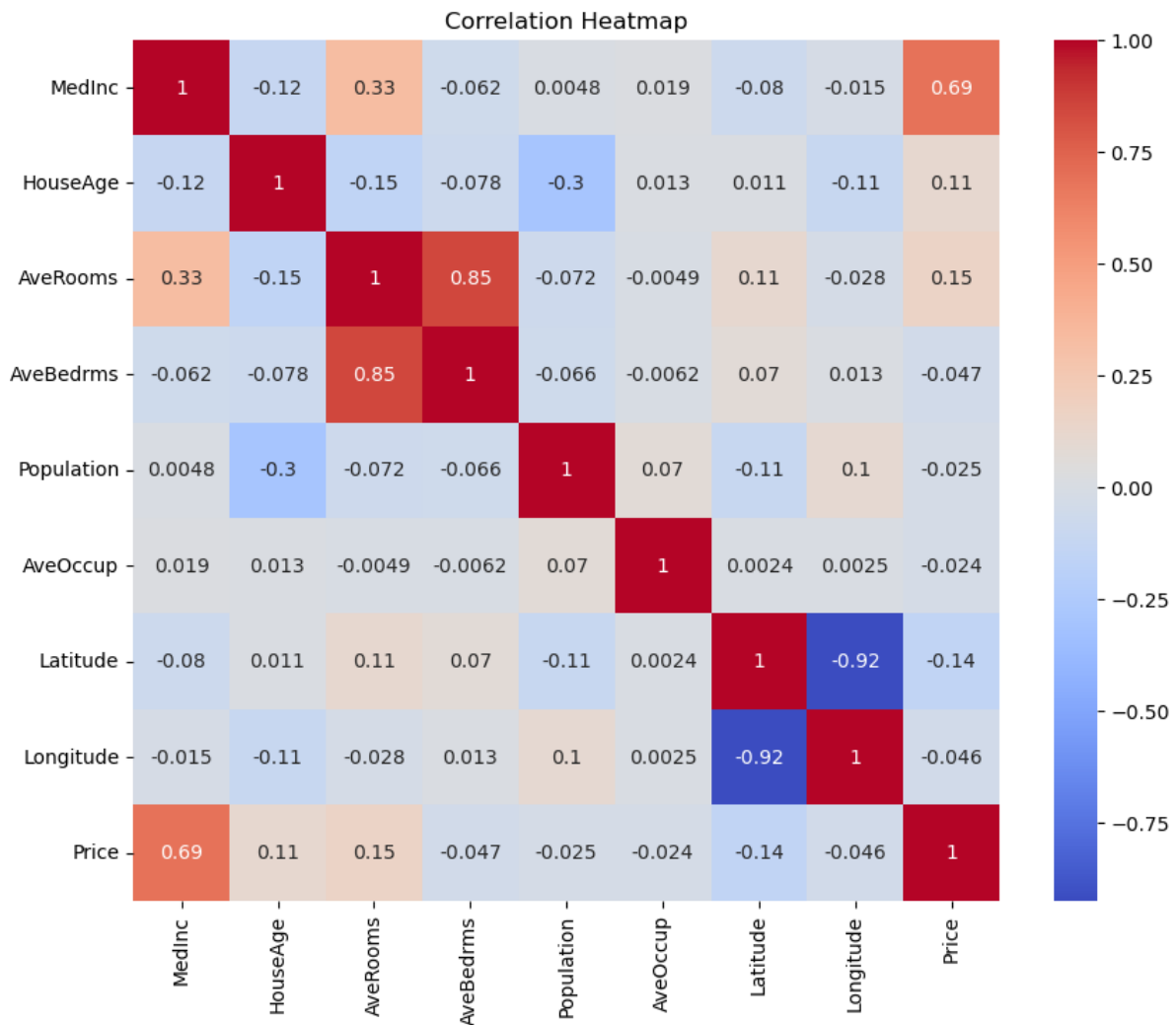


Figure 1: Correlation Heatmap

The heatmap helped visualize how strongly different features are connected with the target variable.

Feature Distribution Histograms

Feature Distribution Histograms

To better understand the distribution of individual features, histograms were created for all important columns in the dataset.

These histograms helped analyze:

- Frequency distribution
- Data spread
- Skewness
- Feature concentration

For example, the Median Income feature showed that most districts have moderate income levels, while very high-income districts are fewer in number.

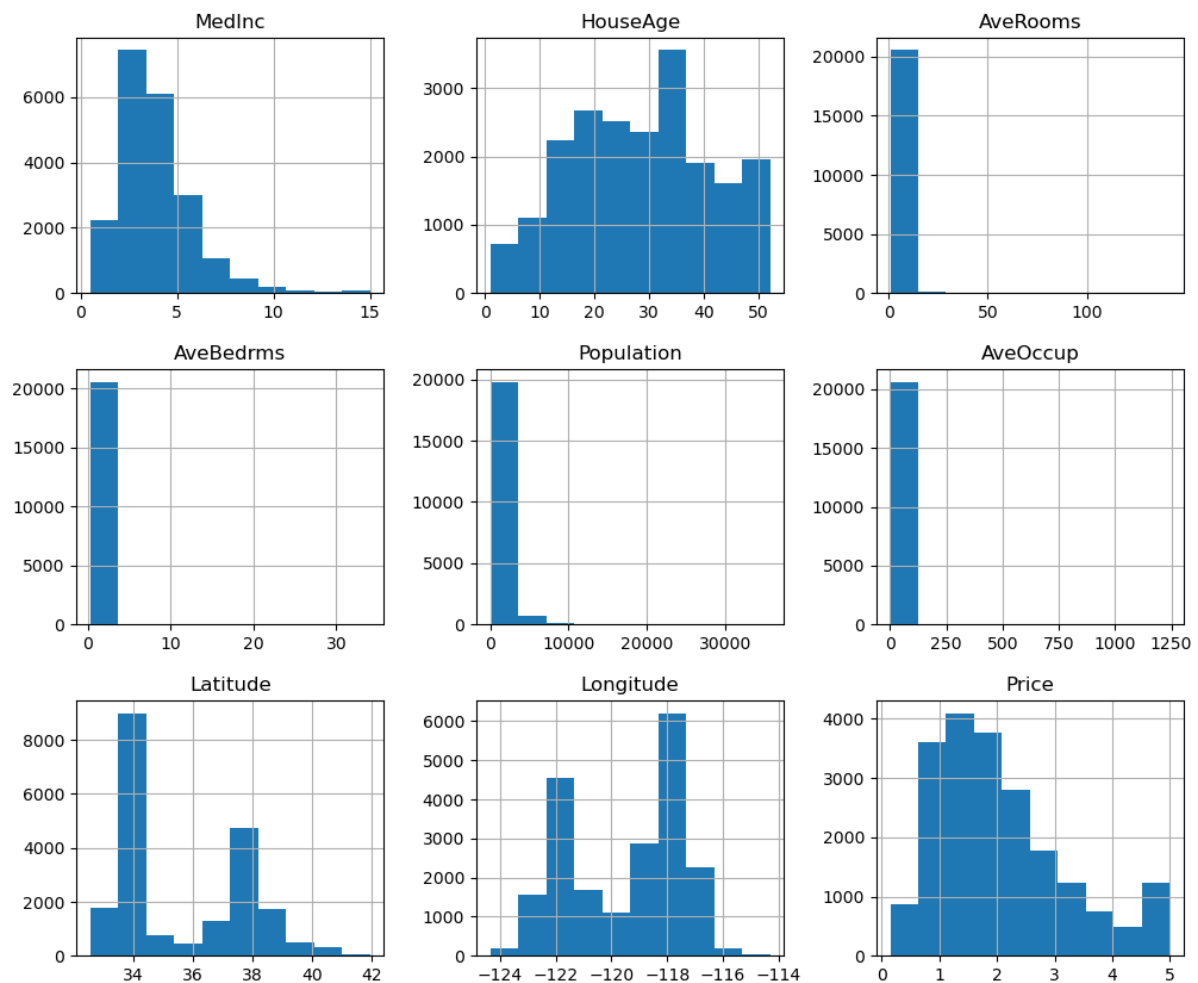


Figure 2: Feature Distribution Histograms

The histograms made it easier to understand how the dataset is distributed and whether any unusual patterns or outliers exist.

Model Training

After completing the data analysis phase, the dataset was divided into training and testing sets using `train_test_split()` from scikit-learn.

The training set contained 80% of the data and was used to train the machine learning model. The testing set contained 20% of the data and was used to evaluate model performance.

A Linear Regression model was then created using scikit-learn. Linear Regression is a supervised machine learning algorithm used for predicting continuous numerical values.

The model learned the relationship between the input features and the target variable during training. Based on these learned patterns, the model could predict house prices for unseen data.

Model Evaluation

After training the model, its performance was evaluated using different regression evaluation metrics.

The following metrics were calculated:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

The results obtained were:

- MAE: 0.533
- RMSE: 0.746
- R^2 Score: 0.576

These values indicate that the model achieved moderate prediction accuracy.

The R^2 score of approximately 0.57 means that the model is able to explain around 57% of the variance in house prices using the available features.

Model Evaluation

Generate

```
mae = mean_absolute_error(y_test, y_pred)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))

r2 = r2_score(y_test, y_pred)

print("MAE:", mae)

print("RMSE:", rmse)

print("R2 Score:", r2)
```

[46] ✓ 0.0s

... MAE: 0.5332001304956558
RMSE: 0.7455813830127765
R2 Score: 0.5757877060324507

Figure 3: Model Evaluation Metrics

The evaluation metrics helped measure how close the predicted house prices were to the actual values.

Visualization

Visualization played an important role in understanding the model performance and interpreting results more clearly.

An Actual vs Predicted scatter plot was created to compare real house prices with the values predicted by the model. Most points followed the diagonal trend, indicating that the model predictions were reasonably close to actual values.

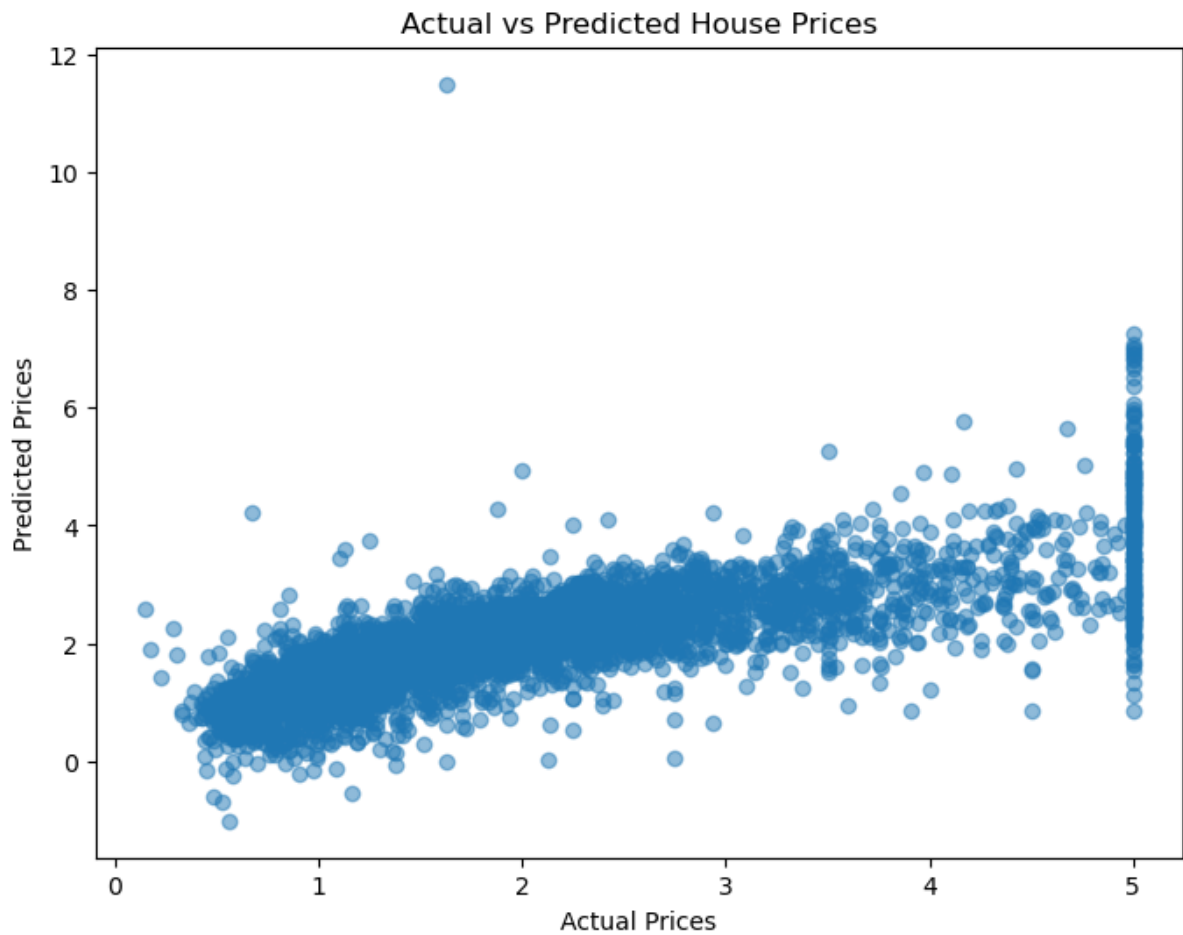


Figure 4: Actual vs Predicted House Prices

The graph shows the relationship between actual and predicted house prices generated by the model.

A Residual Plot was also created to analyze prediction errors. Residuals represent the difference between actual and predicted values.

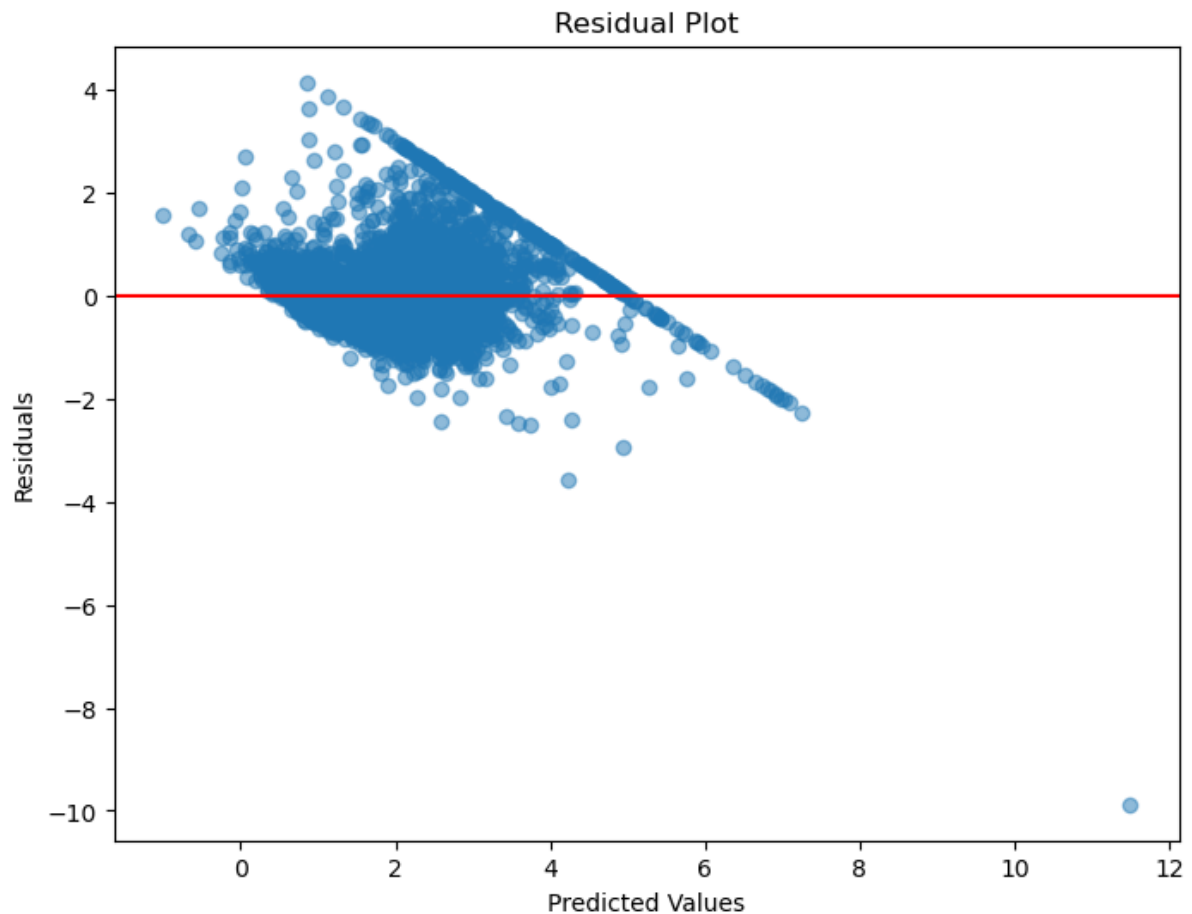


Figure 5: Residual Plot

The residual plot helped identify how prediction errors were distributed across the dataset. Although some variation exists, the errors were generally spread around zero, which indicates acceptable model behavior.

✓ Conclusion

This project successfully demonstrated the complete workflow of a machine learning regression project using Linear Regression and the California Housing dataset.

The model was able to predict house prices with reasonable accuracy. Among all features, Median Income was found to have the strongest influence on house prices.

Through this project, I gained practical experience in:

- Data analysis
- Data visualization
- Machine learning model training
- Model evaluation
- Prediction analysis

This project provided a strong foundation for understanding machine learning concepts and real-world predictive modeling.

Future Improvements

Although the Linear Regression model performed reasonably well, there are several ways to further improve prediction accuracy and overall performance.

Possible future improvements include:

- Feature Engineering
- Polynomial Regression
- Random Forest Regression
- Hyperparameter Tuning
- Advanced preprocessing techniques
- Feature scaling and normalization

More advanced machine learning models may capture complex patterns better and improve prediction performance further.